Energy Efficiency of LLM-Generated Code: A Cross-Language and Cross-Platform Empirical Study

Brisild Capani
[TBD] Affiliation
brisild.capani@aeb.com

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Abstract

TODO: Translate and condense the existing German abstract (~300 words) to ~200 words. Final numbers and findings can only be added after the experiments; until then, the abstract describes goal and method only.

Large Language Models (LLMs) are increasingly used to generate source code, yet their evaluation focuses predominantly on functional correctness. The energy consumed when the generated code is *executed* has received comparatively little attention. This paper presents an empirical study of the energy efficiency of LLM-generated code along three factors: the generating LLM, the target programming language (Python, Java, C++), and the hardware platform on which the code runs (desktop PC, Raspberry Pi, server). Using a full factorial design over [TBD] programming tasks, we measure energy consumption, execution time, and peak memory usage of functionally correct solutions and analyse main and interaction effects with mixed-effects models. In particular, we investigate whether the relative ranking of LLMs with respect to energy efficiency is stable across languages and platforms.

TODO: Results and implications.

1 Introduction

LLM-based assistants have become part of everyday software development, generating anything from small snippets to complete functions. Benchmarks for code generation predominantly measure *functional correctness*, e.g., the share of generated solutions that pass a set of unit tests. Non-functional properties such as runtime performance and, in particular, energy efficiency are rarely considered, even though generated code may be executed many times in production and on a wide range of devices, from data-centre servers to resource-constrained embedded systems.

TODO: Add 1–2 citations on the ICT/software share of energy consumption and on the adoption of AI coding assistants.

Early studies indicate that the energy efficiency of LLM-generated code varies between models and often falls short of human-written reference solutions [1, 2, 4]. However, existing studies typically vary only a subset of the factors that plausibly influence energy consumption: they either cover several languages on a small number of similar platforms [4], or several heterogeneous platforms for a single language [1]. It therefore remains unclear whether a statement such as "model A generates more energy-efficient code than model B" holds in general, or only for a particular language and execution environment.

This paper addresses this question with a full factorial experiment in which [TBD] LLMs generate solutions for [TBD] programming tasks in Python, Java, and C++. All functionally correct solutions are executed on three heterogeneous platforms (an x86 desktop PC, an x86

server, and an ARM-based Raspberry Pi), and their energy consumption, execution time, and memory usage are measured under controlled conditions. Our central research question is:

> *To what extent do the choice of LLM, programming language, and hardware platform influence the energy efficiency of LLM-generated code?*

The contributions of this paper are:

- A full factorial empirical study of LLM-generated code across [TBD] LLMs, three programming languages, and three hardware platforms.
- A statistical analysis of main *and interaction* effects, with a particular focus on whether the energy-efficiency ranking of LLMs is stable across languages and platforms.
- An analysis of the relationship between energy, execution time, and memory for generated code.
- A replication package containing tasks, prompts, generated code, measurement scripts, and raw data.

TODO: Paper outline paragraph ("The remainder of this paper is structured as follows...") once the structure is final.

2 Background

2.1 LLM-Based Code Generation

TODO: Short overview: how LLMs generate code, common benchmarks (HumanEval, MBPP, EvalPlus, EvoEval, LeetCode-based sets), pass@k as correctness metric. Cite originals.

2.2 Software Energy Measurement

TODO: Energy vs. power ($E = \int P\,dt$); software-based estimation (Intel/AMD RAPL, tools building on it such as Scaphandre, CodeCarbon, perf) vs. hardware-based measurement (external power meters, INA219/INA226 shunt sensors). Explain which components each method captures (package, DRAM, whole system) and why RAPL is not available on ARM-based Raspberry Pi boards.

2.3 Energy Efficiency of Programming Languages

Pereira et al. [3] compared 27 programming languages on the Computer Language Benchmarks Game and examined how energy, time, and memory relate.

TODO: Summarise key methodological points relevant to our study (compiled vs. virtual-machine vs. interpreted languages; faster is not always more energy-efficient) – in own words, without anticipating our results.

3 Related Work

Energy efficiency of LLM-generated code. Solovyeva et al. [4] analyse the energy efficiency and performance of code generated by GitHub Copilot, GPT-4o, and o1-mini for "hard" LeetCode problems in Python, Java, and C++ on two platforms (a Mac and a PC). They report that the models are considerably more successful at generating correct Python and Java code than C++ code. Islam et al. [2] evaluate code generated by 20 LLMs for 878 LeetCode problems against canonical human-written solutions and find that LLM-generated solutions are frequently less energy-efficient, with large differences between models and algorithmic categories. Apsan et al. [1]

Table 1: Comparison of closely related studies (based on abstracts; to be verified).

Study	#LLMs	Languages	Platforms	Interaction analysis
Solovyeva et al. [4]	3	Py, Java, C++	Mac, PC	[TBD]
Islam et al. [2]	20	[TBD]	[TBD]	[TBD]
Apsan et al. [1]	6	Py	Server, PC, RPi	[TBD]
This study	[TBD]	Py, Java, C++	PC, Server, RPi	yes

compare Python code generated by six LLMs with four prompting techniques against human-written solutions and those of a green-software expert on nine EvoEval problems, measuring energy on a server, a PC, and a Raspberry Pi. They observe that the relative efficiency of LLM-generated versus human code, as well as the most effective prompting technique, varies with the hardware platform.

Energy efficiency of programming languages. Pereira et al. [3] established that languages differ substantially in energy, time, and memory, and that these three measures are related but not interchangeable.

TODO: Extend the literature search: (i) performance/efficiency of LLM-generated code (e.g., EffiBench, EvalPerf/EvalPlus, Mercury, ECCO); (ii) green prompting / energy-aware code generation; (iii) energy measurement methodology (RAPL validation studies); (iv) updated language-energy studies. Document search strings and databases.

Research gap. Table 1 contrasts the studies along the three factors considered in this paper. To the best of our knowledge, no prior study varies the LLM, the programming language, and a set of architecturally heterogeneous hardware platforms (including an ARM-based, resource-constrained device) in a single factorial design and explicitly tests their interaction effects.

Validate after full literature search.

4 Research Questions and Hypotheses

RQ1 Which LLMs generate the most energy-efficient code for identical programming tasks?

RQ2 How does the energy efficiency of generated code differ between Python, Java, and C++?

RQ3 How do energy consumption and execution time of generated code change across a desktop PC, a Raspberry Pi, and a server?

RQ4 Are there statistically significant interaction effects between LLM, programming language, and hardware platform, i.e., does the ranking of LLMs change depending on language and platform?

RQ5 How are execution time, energy consumption, and memory usage related?

Each research question is accompanied by a hypothesis (null hypotheses $H_{x,0}$ state the absence of the effect). All hypotheses are tested; none is presupposed to hold.

H1 (RQ1)
Energy consumption of generated code differs between LLMs (main effect of LLM).

H2 (RQ2, RQ4)
The effect of the LLM on energy consumption depends on the programming language (LLM $\times$ language interaction).

H3 (RQ3, RQ4)
The effect of the LLM on energy consumption depends on the hardware platform (LLM × platform interaction).

H4 (RQ4)
There is a three-way interaction LLM × language × platform.

H5 (RQ5)
Execution time and energy consumption are correlated, and the strength of this relation differs between languages and platforms.

TODO: Decide whether language and platform main effects (RQ2/RQ3) get their own hypotheses. Note: the language main effect is largely expected from prior work [3]; the novelty lies in the interactions.

TODO: Pre-register hypotheses and analysis plan (e.g., OSF) before data collection.

5 Experimental Setup

5.1 Study Design

We use a full factorial design with the independent variables *LLM* ([TBD] levels), *programming language* (Python, Java, C++), and *hardware platform* (PC, server, Raspberry Pi). Programming tasks are treated as a random factor. The dependent variables are energy consumption (J, primary), execution time, peak memory usage, and functional correctness.

5.2 Large Language Models

TODO: Select models. Criteria: mix of proprietary and open-weight models, different sizes/vendors, availability of a fixed model version/snapshot for reproducibility. Record exact model identifiers, access dates, and decoding parameters.

Table 2: LLMs under study.

Model	Vendor	Version / snapshot	Access
[TBD]	[TBD]	[TBD]	API / local

5.3 Programming Tasks

TODO: Select 10–15 tasks covering sorting, searching, numerical computation, string processing, recursion, data structures, and memory-intensive workloads. Requirements: (i) language-independent specification, (ii) a test suite for correctness, (iii) a scalable input size so that each run lasts long enough for reliable energy measurement (target: ≥[TBD] s per run), (iv) low risk of training-data contamination. Candidate sources: Computer Language Benchmarks Game, EvoEval, EvalPerf, adapted LeetCode/Rosetta Code problems.

5.4 Hardware Platforms and Software Environment

TODO: Use identical language-runtime versions on all platforms where possible (e.g., via a pinned container image or toolchain). Fix CPU frequency governor, disable turbo/boost where feasible, and document all deviations.

Table 3: Hardware and software configuration.

	Desktop PC	Server	Raspberry Pi
CPU / architecture	[TBD]	[TBD]	[TBD] (ARM)
RAM	[TBD]	[TBD]	[TBD]
OS / kernel	[TBD]	[TBD]	[TBD]
Python (interpreter)	[TBD]	[TBD]	[TBD]
Java (JDK / JVM)	[TBD]	[TBD]	[TBD]
C++ compiler, flags	[TBD]	[TBD]	[TBD]
Energy measurement	[TBD]	[TBD]	[TBD]

5.5 Energy Measurement

TODO: Decide measurement instruments per platform. Key issue: RAPL (PC, server) measures CPU package/DRAM, whereas an external meter (Raspberry Pi) measures whole-board power. Options: (a) use external whole-system meters on all three platforms for comparability, optionally complemented by RAPL; (b) compare rankings *within* each platform rather than absolute Joules *across* platforms. Subtract idle baseline power.

6 Methodology and Data Collection

6.1 Prompting and Code Generation

TODO: Define the prompt template; it differs between languages only in the target language and the required I/O interface. Specify: zero-shot, no energy-related hints (baseline), temperature, number of samples per task/LLM/language (k), handling of non-compiling outputs (no manual repair, or a fixed number of automated retries).

6.2 Functional Correctness

TODO: Every generated program is executed against the task's test suite. Only programs passing all tests enter the energy analysis. Correctness rates are reported separately per LLM and language (**RQ1**, **RQ2**).

6.3 Measurement Procedure

TODO: Describe per run: warm-up, number of repetitions (n), randomised execution order, cool-down / idle intervals, idle-power baseline, JVM warm-up policy, process isolation, minimal background services. Record ambient temperature if possible.

6.4 Statistical Analysis

TODO: Planned analysis (to be finalised before data collection): (i) descriptive statistics (median, IQR) per cell; (ii) linear mixed-effects model on log(energy) with fixed effects LLM, language, platform and all interactions, and random intercepts for task (and generated solution); (iii) Type-III tests / likelihood-ratio tests for interactions, effect sizes, post-hoc comparisons with multiplicity correction; (iv) rank stability across languages and platforms via Kendall's τ / Friedman test (**RQ4**); (v) Spearman correlation between energy, time, and memory (**RQ5**). Check model assumptions; fall back to non-parametric or aligned-rank-transform methods if violated.

7 Results

7.1 Functional Correctness

TODO: Correctness rates per LLM $\times$ language.

7.2 RQ1: Differences between LLMs

TODO:

7.3 RQ2: Differences between Programming Languages

TODO:

7.4 RQ3: Differences between Hardware Platforms

TODO:

7.5 RQ4: Interaction Effects and Rank Stability

TODO:

7.6 RQ5: Relation of Energy, Time, and Memory

TODO:

8 Discussion

TODO: Interpret results with respect to the RQs; relate to prior work [1–4]; analyse *why* differences arise (e.g., algorithmic choices, library usage, data structures) by qualitatively inspecting selected solutions; implications for developers and for LLM benchmarking.

9 Threats to Validity

Construct validity. Energy measurement methods differ in which components they capture (CPU package and DRAM via RAPL vs. whole-board power via external meters), which limits the comparability of absolute values across platforms.

TODO: Describe mitigation chosen in Section 5.

Internal validity. Measurements may be influenced by background processes, thermal state, frequency scaling, JIT warm-up, and garbage collection. LLM outputs are non-deterministic.

TODO: Mitigations: repetitions, randomised order, fixed governor, multiple samples per task.

External validity. Results are limited to the selected LLMs (and their versions at the time of the study), tasks, three languages, and three platforms, and may not generalise to other models, languages, or architectures, or to larger, real-world codebases.

Conclusion validity.

TODO: Multiple testing, assumption violations, sample size / power.

Confounding of model and implementation. The effect of an LLM cannot be fully separated from the concrete algorithm it chooses; these strategy differences are, however, part of the phenomenon under study.

Data contamination. Benchmark tasks and reference solutions may be part of the LLMs' training data.

10 Conclusion and Future Work

TODO: Summarise answers to RQ1–RQ5 once results are available.

Future work.

TODO: E.g., energy-aware prompting, additional languages (Rust, Go, JavaScript), GPU workloads, larger real-world tasks, energy cost of generation vs. execution.

Data availability.

TODO: Link to replication package (tasks, prompts, generated code, scripts, raw data).

References

[1] Radu Apsan, Vincenzo Stoico, Michel Albonico, Rudra Dhar, and Karthik Vaidhyanathan. Generating energy-efficient code via large-language models – where are we now? arXiv preprint arXiv:2509.10099, 2025. URL `https://arxiv.org/abs/2509.10099`.

[2] Md Arman Islam, Devi Varaprasad Jonnala, Ritika Rekhi, Pratik Pokharel, Siddharth Cilamkoti, Asif Imran, Tevfik Kosar, and Bekir Turkkan. Evaluating the energy-efficiency of the code generated by LLMs. arXiv preprint arXiv:2505.20324, 2025. URL `https://arxiv.org/abs/2505.20324`.

[3] Rui Pereira, Marco Couto, Francisco Ribeiro, Rui Rua, Jácome Cunha, João Paulo Fernandes, and João Saraiva. Energy efficiency across programming languages: How do energy, time, and memory relate? In *Proceedings of the 10th ACM SIGPLAN International Conference on Software Language Engineering (SLE)*, pages 256–267, New York, NY, USA, 2017. ACM. doi: 10.1145/3136014.3136031.

[4] Lola Solovyeva, Sophie Weidmann, and Fernando Castor. AI-powered, but power-hungry? Energy efficiency of LLM-generated code. arXiv preprint arXiv:2502.02412, 2025. URL `https://arxiv.org/abs/2502.02412`.

Open TODOs

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